Assignment week 10: Design for change

Options for business cases (with descriptions):

1. **Library management system (example for inspiration)**:
    - **Description**: This system manages books, students (members), librarians and lending activities in a university library. Students can search for available books, borrow them and return them. Librarians can manage the book stock and handle loan requests.
    - **Examples of concepts**: book, student, librarian, borrowing, reserving.
    - **Use cases**: Borrowing a book, returning a book, searching the catalogue, reserving a book.
    - **Domain objects**: book, student, librarian, loan transaction.
2. **Online food ordering system**:
    - **Description**: This system allows customers to view menus from different restaurants, place orders and track deliveries. Restaurant owners can manage their menus and delivery drivers can update the status of deliveries.
3. **Student registration system**:
    - **Description**: This system allows students to register for courses, view timetables, and manage study plans. Lecturers can manage course offerings and view registration lists.
4. **Parking management system**:
    - **Description**: This system manages parking spaces for an office complex. Employees can reserve parking spaces, check availability, and manage their reservations. Administrators can allocate parking spaces and ensure efficient use of the parking spaces.
5. **Online ticket sales system for events**:
    - **Description**: This platform allows users to purchase tickets for events such as concerts, sports matches and theatre performances. Event organisers can post events and users can view event details and reserve tickets.

Part 1: Group assignment Visibility

Apply this week's theory to the business case you chose at the beginning of this academic year. Expand the DCD with visibility.

Part 2: Group assignment GRASP

Apply this week's theory to the business case you chose at the beginning of this academic year. Apply GRASP to the SDs from two weeks ago and also to the DCD.

Part 3: Individual research task – Abstract classes & interfaces in a DCD

In preparation, each student will conduct individual research on **Abstract classes & interfaces in a DCD**.

Answer the following questions in a short summary:

- What are abstract classes?
- What are interfaces?
- What techniques does it contain?
- What is its purpose in software development?

To be submitted individually:

- A short summary of your findings (±200 words).